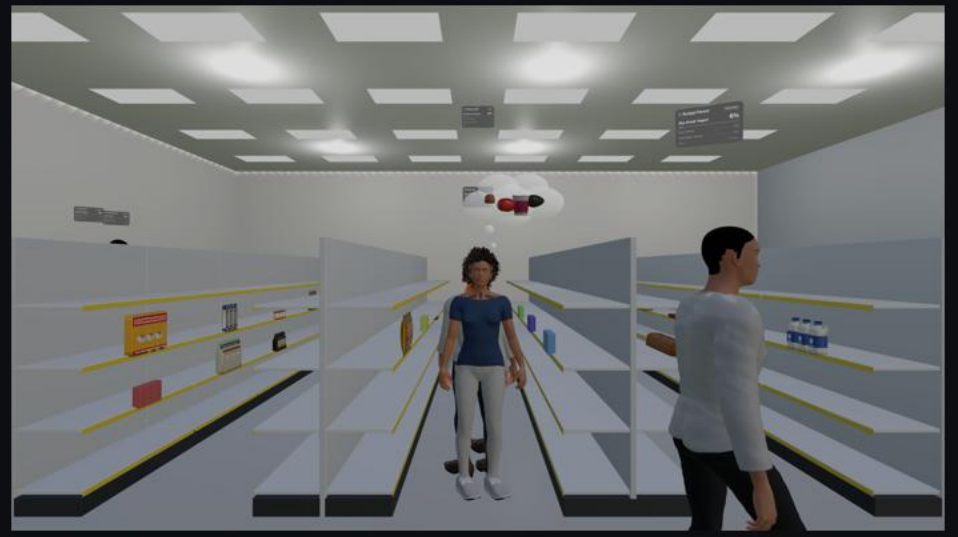
Behavior Aisle demo plan

Sundai Hack 148 · Behavioral Models · 8 Nov 2026

Coverage of expected features, must-finish list, 3-minute run sheet.



Sundai Hack 148, MIT, 8 Nov 2026

Theme: Behavioral Models, predict the next action (Unbox AI BehaviorGPT).
Source: PRD.md. sundai.club event pages load client-side and /hacks needs sign-in; theme, sponsors and judging rubric were not readable by fetch on 2026-11-08.

What judges of a behavior-model hack check:

1. The model is live and visibly drives behavior (not scripted).
2. Next action is legible: item, probability, and why.
3. A change to the world changes the prediction (cause -> effect).
4. Some proof it is right: accuracy, hit rate, or a baseline comparison.
5. A buyer and a use case: who pays, what decision it replaces.

Expected features vs built

BUILT	Live BehaviorGPT next-action calls via proxy	Predict/BehaviorGPTPredictor.swift, proxy/
BUILT	Offline fallback: replay log, then heuristic	RoutedPredictor
PARTIAL	Lifesize store with stocked shelves	StoreBuilder + StoreDensity (uncommitted)
BUILT	Shoppers act on predictions in a loop	AppModel.request/execute/finishPick
BUILT	Top-3 next actions with probability	PredictionCard, IntentPanel
BUILT	Reason per prediction (interest -> item)	IntentExplainer
BUILT	Destination shown in space: path, beacon	Viz/IntentLayer.swift
BUILT	Branch view: ghosts walk top 3	Viz/GhostLayer.swift
BUILT	What-if: move product, sale sign, re-query	moveProduct, toggleSale
BUILT	Store heatmap	Viz/HeatmapLayer.swift
BUILT	Anomaly flag at checkout	anomaly badge
BUILT	Real person in: X profile -> persona	Profile/, ProfileImportView
BUILT	Basket timeline + top-1 hit counter	IntentPanel.BasketTimeline (this pass)
BUILT	Before/after metric for a what-if	Experiment.whatIf, paired 95% CI
BUILT	Live vs heuristic side by side	IntentPanel.BaselineRow
BUILT	Accuracy vs baselines + ablations	Experiment.accuracy (simulated truth)
PARTIAL	Run on device / simulator end to end	PRD status: not yet run

15/17 built

Must finish before demo

- **P0** Run the full demo route on simulator once; fix any crash.
  demo blocker
- **P0** Confirm Unbox field names in proxy/worker.js at workshop; one live call.
  theme fit
- **P0** Commit StoreDensity + intent layer; restock to packed preset.
  store reads finished
- **P1** What-if delta chip: 'Paper towels +38% picks' after sale toggle.
  proof of effect
- **P1** Source toggle on one shopper: BehaviorGPT vs heuristic top 3.
  model value
- **P1** Bundle predictions.jsonl from a live run so replay works offline.
  wifi risk
- **P2** Personal heatmap for the imported profile.
  nice to have
- **P2** Record 60 s backup video via /vision-record-demo.
  fallback

Ideal 3-minute demo

0:00	**Hook**	A shelf move costs weeks of sales data. We predict it in minutes.
0:15	**Real person**	Paste an X handle. Interests -> persona -> likely basket.
0:45	**Enter store**	Shopper walks in; panel shows top 3 next actions, %, reason.
1:05	**Destination**	Floor path and beacon on the rank-1 shelf; shopper walks it.
1:25	**Branches**	Tap shopper: 3 ghosts walk the 3 futures with probabilities.
1:45	**What-if**	Sale sign on paper towels; Budget Parent re-routes live.
2:10	**Proof**	Basket timeline, Top-1 hits 7/9; heatmap over 50 shoppers.
2:40	**Ask**	Merchandisers test planograms on predicted shoppers first.

Fallback: replay mode + bundled predictions.jsonl if wifi or proxy fails.